# Yue Zhao


CONTACT INFORMATION

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RESEARCH SUMMARY

When something goes wrong, someone has to be able to say what happened and who is answerable. AI systems that act on their own can make that hard: the damage shows up, the path to it often does not. My lab builds three things: tools that catch unsafe permissions before an agent ships, a checkpoint that inspects tool calls before they run, and records good enough to piece together afterward who did what. This work spans **auditable AI systems** across four directions: *auditing of action*, *verification of output*, *detection in data*, and *modeling of risk*. The first three ask whether something departed from what was expected, whether anyone can still tell, and whether the work was necessary at all. Modeling of risk, a developing direction, asks what that evidence can establish about future losses in a given deployment. Beyond a system's behavior, my lab studies its real-world impact: how LLMs and agents can affect people and the decisions made with them. The current frontier is **agent auditability**: deployable open-source infrastructure that audits agents from before deployment to after a run.

### 1. Auditing of Action: Auditability and Control of Agent Systems

When a system acts through tools, APIs, and shared state, what it does becomes the primary failure surface. This thread covers auditability frameworks, runtime control surfaces, agent-specific failure modes, and post-run attribution and recovery, spanning the full lifecycle from pre-deployment to after a run. It also studies how defense training affects an agent's ability to finish a task, and how workflows can be built and adjusted during a run.

- ❑ Agent Auditability
- ❑ Agent Systems
- ❑ Multi-Agent Systems
- ❑ Behavior Verification
- ❑ AI Agent Security
- ❑ Runtime Guardrails
- ❑ Policy Enforcement
- ❑ Agent Efficiency

### 2. Verification of Output: Trust and Robustness of Foundation Models

Between the input and the action, foundation models add their own failure surface: behavior under attack, hallucination, and adversarial robustness of retrieval. Failures on these surfaces can propagate downstream when model output is passed to an agent or a user-facing system. My lab measures these surfaces through jailbreak detection for vision-language models, causal analysis of hallucination, retrieval-augmented generation attacks, and LLM-as-anomaly-detector benchmarks. Choosing which model answers is part of the same question, treated as an online cost-quality decision.

- ❑ Foundation-Model Trust
- ❑ Trustworthiness Benchmarks
- ❑ LLM Hallucination
- ❑ Jailbreak Detection
- ❑ RAG Security
- ❑ Adversarial Robustness
- ❑ Multimodal Trust

### 3. Detection in Data: Anomaly and Out-of-Distribution Detection

Here the question is whether an input departed from the conditions represented during development and evaluation. This is my methodological and open-source base, built through the PyOD ecosystem and ADBench. It continues through benchmarks for NLP, automatic OOD detector selection, modality-specific OOD methods for graphs, multimodal data, and video, and few-shot cross-domain OOD detection. When an incident involves distribution shift, these methods supply evidence about whether unfamiliar input conditions contributed to it. Large-scale detection systems cut the compute

cost of running many detectors.

- ❑ Anomaly Detection
- ❑ Outlier Detection
- ❑ Out-of-Distribution Detection
- ❑ Distribution Shift
- ❑ Multimodal OOD
- ❑ Graph OOD
- ❑ Benchmark Design

**4. Modeling of Risk: Agent Risk, Loss, and Decision-Making**
This developing direction connects behavioral evidence from the other three threads to the likelihood and severity of losses in a specified deployment. It accounts for permissions, controls, exposure, and the time horizon. The thread studies how risk changes as agents, tools, and workflows evolve, and how estimates could inform deployment decisions, risk controls, and insurance underwriting. Losses include harm from a wrong action and the cost of work that was not needed.

- ❑ Agent Risk Modeling
- ❑ Loss Estimation
- ❑ Dynamic Risk Assessment
- ❑ Deployment Decisions
- ❑ Insurance Underwriting

Tools from this group have been cited in a U.S. Senate committee report, a NIST special publication, and the U.S. Department of Defense Generative AI Toolkit. PyOD was selected by the European Space Agency for spacecraft anomaly detection and is the subject of five published books. TrustLLM serves as an official benchmark in all four editions of the Future of Life Institute AI Safety Index through Summer 2026.

## Professional Experience

**University of Southern California**
*Thomas Lord Department of Computer Science*
Assistant Professor (Tenure-Track) — Aug. 2023 - Present

- Foundations Of Robust Trustworthy Intelligent Systems (**FORTIS**) Lab: Link
- USC Machine Learning Center (MaSCle): Link

**Fig Work**
*AI-powered marketplace for early-career talent*
Chief Scientific Advisor — Jun. 2026 - Present

**PwC Canada**
*Consulting & Deals*
Senior Consultant (Data Scientist) — Aug. 2017 - Jun. 2019
Consultant (Data Scientist) — Feb. 2017 - Jul. 2017
Research Associate (Intern) — May. 2016 - Jan. 2017

## Education

**Carnegie Mellon University** — Pittsburgh, PA
*Ph.D. in Information Systems and Management* — Sep. 2019 - May. 2023
Heinz College of Information Systems and Public Policy

- **Affiliation**: CMU automated learning systems group (Catalyst) and Data Analytics Techniques Algorithms (DATA) Lab
- **Advisors and Mentors**: CMU: Prof. Leman Akoglu, Prof. Zhihao Jia, and Prof. George Chen. I collaborate with Prof. Jure Leskovec at Stanford, and Prof. Philip S. Yu at UIC.

**University of Toronto** — Toronto, ON
*Master of Science in Computer Science* — Sep. 2015 - Dec. 2016

**University of Cincinnati** — Cincinnati, OH
*Bachelor of Science in Computer Engineering* — Sep. 2010 - May. 2015
**Minor**: *Computer Science* and *Mathematics*

## Awards, Grants, and Funding

**Since August 2023**

| | | |
|---|---|---|
| Survival and Flourishing Fund (SFF), Speculation Grant | *Grant* | Oct. 2026 |
| NSF FIRE (co-PI) | *Funding* | Aug. 2026 |
| Meta | *Sponsored Research* | Aug. 2026 |
| Foresight Institute AI for Safety & Science Nodes | *Gift* | Apr. 2026 |
| NVIDIA Academic Grant Program | *Gift* | Mar. 2026 |
| Amazon Research Awards, Fall 2025 | *Gift* | Mar. 2026 |
| USC CCSL Seed Funding | *Small Grant* | Jan. 2026 |
| Second Prize CCC Award @ IEEE ICDM, BlueSky Track | *Recognition* | Nov. 2025 |
| Best Short Paper Award @ ACM SIGSPATIAL | *Recognition* | Nov. 2025 |
| NSF Pathways to Enable Open-Source Ecosystems (POSE), Phase I (co-PI) | *Funding* | Aug. 2025 |
| Capital One Research Awards | *Grant* | Oct. 2024 |
| Amazon Research Awards, Spring 2024 (co-PI) | *Gift* | Aug. 2024 |
| Best Paper Award @ KDD Resource-Efficient Learning Workshop | *Recognition* | Aug. 2024 |
| NSF ATD (co-PI) | *Funding* | Aug. 2024 |
| NSF POSE Phase II (co-PI) | *Funding* | Jun. 2024 |
| Google Cloud Research Innovators | *Recognition* | Mar. 2024 |
| AAAI New Faculty Highlights | *Recognition* | Feb. 2024 |

**Prior to Principal Investigator Role (Before August 2023)**

| | | |
|---|---|---|
| Meta 2022 AI4AI Research Award (student co-PI) | *Recognition* | Oct. 2022 |
| The Norton Labs Graduate Fellowship | *Fellowship* | Mar. 2022 |
| CMU Presidential Fellowship | *Fellowship* | 2019 |
| Mitacs-Accelerate Research and Development Funding | *Funding* | 2016-2017 |
| University Global Award and Scholarship | *Scholarship* | 2010-2015 |
| Mantei/Mae Award & Scholar | *Award* | 2012-2015 |
| Engineer of the Month | *Recognition* | Jun. 2014 |

*Note: Monetary values are omitted for awards and recognitions received prior to PI role.*

## Selected Open-Source Software, Benchmarks, and Systems

Representative open-source research artifacts spanning anomaly detection, trustworthy AI, agent security, and research tooling.

### Primary Contributions

- **PyOD** (*Library*; updated Oct. 2026; 10.1K stars) PyOD 3 is the most comprehensive Python library for anomaly detection: 60+ detectors across tabular, time series, graph, text, and image data, with an ADEngine orchestration core and an agentic investigation layer driven through natural language. 59M+ downloads. Links: docs — paper.
- **Anomaly-Detection-Resources** (*Resource Hub*; updated Mar. 2026; 9.4K stars) A curated resource hub for anomaly detection, including papers, benchmarks, tutorials, datasets, tools, and community references.
- **CS-Paper-Checklist** (*Research Toolkit*; updated May. 2026; 1.6K stars) A practical sanity checklist for improving CS paper quality, structure, and submission readiness.
- **PyGOD** (*Library*; updated Nov. 2024; 1.5K stars) A Python library for graph outlier detection, with unified APIs and broad coverage of graph anomaly detection algorithms. Links: docs — paper.
- **ADBench** (*Benchmark*; updated Jan. 2026; 1.0K stars) Official benchmark for anomaly detection with large-scale evaluations across unsupervised, semi-supervised, and supervised settings. Links: paper.
- **agent-style** (*Developer Toolkit*; updated Oct. 2026; 734 stars) A drop-in rules package that shapes how AI coding and writing agents produce prose at generation time, using 21 curated English style rules (12 from canonical writing authorities; 9 from field observation of LLM output). Works with Claude Code, Codex, Copilot, Cursor, Aider, and other AGENTS.md-compliant tools. Links: docs — demo.
- **combo** (*Library*; updated Sep. 2026; 659 stars) A Python toolbox for machine learning model and score combination across classification, clustering, and anomaly detection tasks. Links: docs — paper.

- **SUOD** (*Library / System*; updated Sep. 2026; 395 stars) An acceleration framework for large-scale unsupervised heterogeneous outlier detection with efficient training and inference. Links: docs — paper.
- **anywhere-agents** (*Developer Toolkit*; updated Nov. 2026; 262 stars) One config to rule all your AI agents: portable across every project and session, effective through curated writing, routing, and skills, and safer via a PreToolUse guard that blocks destructive Git and GitHub commands. Supports Claude Code and Codex today, with plans to grow. Links: docs.
- **awesome-auditable-ai** (*Resource Hub*; updated Oct. 2026; 158 stars) A curated list of papers, tools, datasets, benchmarks, and standards for building, evaluating, and auditing reliable AI agents, organized around a reader-first reliability map.
- **AD-AGENT** (*Platform*; updated Apr. 2026; 105 stars) An LLM-driven multi-agent anomaly detection platform for end-to-end workflows across tabular, graph, and time-series settings. Links: docs — paper.
- **AD-LLM** (*Benchmark*; updated Oct. 2025; 44 stars) A benchmark for using large language models in anomaly detection, covering zero-shot detection, data augmentation, and model selection. Links: paper.
- **auditable** (*Agent Auditability*; updated Oct. 2026; 32 stars) An open-source system of record for AI-agent decisions: it captures what each decision relied on, replays it against live state and rolls back the committed action when it no longer holds, and ranks the steps of a finished run by structural blast share, on one typed graph across an agent's plan, run, and review. Links: docs — paper — demo.
- **agentpile** (*Data Infrastructure*; updated Nov. 2026; 32 stars) A Python package that reads 13 public agent trajectory datasets, trace files, and live agent runs into ATIF v1.7 records that keep what the source recorded, then hands them to pandas, networkx, or OTLP tracing tools. Links: docs.
- **NLP-ADBench** (*Benchmark*; updated Oct. 2025; 21 stars) A benchmark for anomaly detection in NLP, including NLPAD datasets and standardized evaluation across end-to-end and embedding-based methods. Links: paper.
- **CatchBench** (*Benchmark*; updated Oct. 2026; 13 stars) A benchmark for auditing agent failures across the full lifecycle. It scores one auditor's question against three information states: the declared configuration before a run, a growing prefix of the trace, and the finished trace. The release covers 72 entrants and 1187 declared configurations. Contrasts that do not separate are published unresolved rather than ranked. Links: paper.

### Selected Collaborations

- **TODS** (*System / AutoML*; updated Sep. 2023; 1.7K stars) An automated time-series outlier detection system with end-to-end pipeline support for preprocessing, feature extraction, model search, and detection. Links: docs — paper.
- **TDC** (*Benchmark / Platform*; updated Jul. 2025; 1.3K stars) Therapeutics Data Commons, a platform for AI-ready therapeutic datasets, benchmarks, and evaluation workflows across drug discovery tasks. Links: docs — paper.
- **TrustLLM** (*Benchmark / Toolkit*; updated Oct. 2026; 635 stars) A trustworthiness benchmark and toolkit for LLMs; collaboration included follow-on NSF-supported development with the core team. Links: docs — paper.
- **Aegis** (*Security / Compliance Platform*; updated Oct. 2026; 524 stars) The open-source firewall for AI agents, with pre-execution policy checks, human-in-the-loop approvals, and tamper-evident audit trails for safer and more accountable deployment. Links: docs — paper — demo.
- **agent-audit** (*Security Toolkit*; updated Sep. 2026; 245 stars) An open-source tool for auditing AI agent code and configurations, with OWASP Agentic Top 10 coverage, taint analysis, and MCP configuration auditing. Links: docs — paper — demo.
- **TrustEval-toolkit** (*Benchmark / Toolkit*; updated Oct. 2026; 136 stars) A toolkit and benchmark for evaluating trustworthiness of generative foundation models across LLM, VLM, and text-to-image settings. Links: docs — paper.
- **LangSkills** (*Toolkit / Knowledge Base*; updated Mar. 2026; 116 stars) An evidence-backed skill library and pipeline for AI agents, with offline searchable bundles distilled from papers and technical sources.

## Preprints & Under Submission

Note: †Equal contribution.

137. Bingran You†, Wenjun Ke†, Ria Rosenauer†, Bo Liu†, Wenbo Chen, Min Chen, Shuqi Xu, Yue Ma, Jiechao Feng, Bo Chen, Xiangkai Sun, Jiarui Liu, Jinen Guo, Zeyu Zhou, Xu Huang, Zhuohan Li, Haoyu Li, Heguang Lin, Lijie Ding, Yadong Zeng, Kangqi Zhang, Xinge Liu, Hanwen Liu, Zhenghong Li, Alexander von Recum, Yuchen Zhu, Felix Schwarzfischer, Yizhou Zhao, Jicheng Wang, Binxu Li, Qian Yu, Chan Li, Cheng Dai, Di Wang, Haoyu Zhang, Jiankai Sun, Kaituo Feng, Mian Qin, Ruihong Yin, Yiqun Luo, Weiwei Wu, Shihong Pan, Tsung Yeh Hsieh, Xiao Huang, Yan Zhang, Sihan Li, Zirui Ren, Zengji Tu, Kyoung Whan Choe, Ke Sun, Vincenzo Galia, Vaisakhi Mishra, Jiashu He, Hongyang Wang, Zhiheng Yang, Shubham Gaur, Yifan Gu, Shuyi Wang, Zihao Zuo, Steven Dillmann, Yiyou Sun, Xinyang Han, Hanchung Lee, Molei Tao, Yue Zhao, Shilong Liu, Yuchen Wu, Xuandong Zhao, Xiangyi Li, Florian Brand, Gerbrand Ceder, Hartmut Haeffner
FrontierPhysics: Evaluating Agents for End-to-End Physics Research
**arXiv preprint arXiv:2611.06910**

136. Yu Jiang†, Yue Huang†, Hang Hua, Yuyang Wu, Zixuan Weng, Olexandr Isayev, Yue Zhao, Zinan Lin, Xiangliang Zhang
Is Your Long-Horizon Benchmark Really Deep? Measuring and Synthesizing Depth in Terminal Tasks
**arXiv preprint arXiv:2611.01927**

135. Haoming Li†, Defu Cao†, Xu Hu†, Yingtao Luo, Yiyang Wang, Xianghe Xu, Jiashu He, Wanting Jiang, Yifan Zhang, Yunhan Zhao, Siyang Wu, Chen Yang, Haixin Wang, Zheyuan Deng, Taoran Li, Jingwen Zhang, Zhejian Peng, Shushu Liang, Yue Zhao, Bingzhe Li, Feng Chen, Yifan Wang, Jintao Huang, Yijun Wang, Yuexing Hao, Xiaomin Li
PersonaCUABench: Benchmarking Computer-Use Agents Serving Diverse Personas
**arXiv preprint arXiv:2610.22990**

134. Zixiang Xu, Zirui Song, Wanrong Zheng, Chiyu Zhang, Yue Huang, Yuyang Peng, Kexin Zhang, Jinkun Chen, Xiuying Chen, Xiangliang Zhang, Yan Liu, Laurent Itti, Frank Shyu, Xiaoqiang Luo, Sandeep Pandey, Xi Liu, Luke Simon, Xiyang Hu, Yue Zhao
Grid-Bench: Can LLMs Predict How Discrete Worlds Change?
**arXiv preprint arXiv:2610.22312**

133. Yue Zhao, Zixiang Xu, Angela Huyue Zhang, S. Alex Yang, Xiyang Hu
agent-risk: Risk Assessment for Changes to AI Agents
**arXiv preprint arXiv:2610.16280**

132. Yue Zhao, Zuobin Xiong, Xiyang Hu, Zhangyu Wang, Xiongye Xiao, Ruolin Li
AI4Fire: Evaluating Large Language Models on Wildfire Tasks
**arXiv preprint arXiv:2610.10946**

131. Yue Zhao, Jiate Li, Li Li, Yi Nian, Jinbo Liu, Xiaolin Zhou, Xiyang Hu
Auditable Claims about AI Agents
**arXiv preprint arXiv:2610.07459**

130. Jinyi Ye, Yuangang Li, Chenxiao Yu, Preyashi Poddar, Priyanka Dey, Longtian Ye, Zihan Wang, Xiyang Hu, Emilio Ferrara, Yue Zhao
MetaPersona: Task-Grounded Synthetic Populations from Empirical Social Science
**arXiv preprint arXiv:2609.38392**

129. Jesson Wang†, Shawn Li†, Wei Yang, Franck Dernoncourt, Ryan A. Rossi, Charith Peris, Yue Zhao
Controlled Decoding Attacks on Black-Box LLMs
**arXiv preprint arXiv:2609.36956**

128. Prince Zizhuang Wang†, Chenhao Liang†, Zelong Xu†, Aojie Yuan†, Xiaolin Zhou†, Haiyue Zhang, Yue Zhao, Xiyang Hu, Shuli Jiang
CUA-SWE: When Computer-Use Agents Meet Visual Software Engineering
**arXiv preprint arXiv:2609.32600**

127. Yue Zhao, Huayue Gu, Yushun Dong, Xiyang Hu
QEMScore: How Much Does the Measurement Add to Learned Quantum Error Mitigation?
**arXiv preprint arXiv:2609.17896**

126. Yue Zhao, Mengyuan Li, Ruolin Li, Prince Zizhuang Wang, Shuli Jiang, Linsey Pang, Xiongye Xiao, Xiyang Hu
CatchBench: When Can an Agent Failure Be Caught?
**arXiv preprint arXiv:2608.22808**

125. Shawn Li, Wei Yang, Jike Zhong, Jiate Li, Jiawei Yang, You Qin, Ryan Rossi, Franck Dernoncourt, Roger Zimmermann, Yue Wang, Zhengzhong Tu, Vicente Ordonez, Mohit Bansal, Yue Zhao
JigShape: Evaluating Visual-Geometric Reasoning in VLMs through Jigsaw Puzzles
**arXiv preprint arXiv:2607.27670**

124. Yue Huang, Wenjie Wang, Han Bao, Yuchen Ma, Xiaonan Luo, Yi Nian, Haomin Zhuang, Zheyuan Liu, Yue Zhao, Xiangliang Zhang
MemoHarness: Agent Harnesses That Learn from Experience
**arXiv preprint arXiv:2607.14159**

123. Tianming Sha, Yue Zhao, Lichao Sun, Yushun Dong
SkillCenter: A Large-Scale Source-Grounded Skill Library for Autonomous AI Agents
**arXiv preprint arXiv:2607.07676**

122. Shawn Li, Yue Zhao
Agent Safety Is Action Alignment
**arXiv preprint arXiv:2606.28739**

121. Yue Zhao
GRADE: Graph Representation of LLM Agent Dependency and Execution
**arXiv preprint arXiv:2606.22741**

120. Ojas Nimase, Jiate Li, Yue Zhao, Yushun Dong
Can Subgraph Explanations Be Weaponized to Steal Graph Neural Networks?
**arXiv preprint arXiv:2605.30470**

119. Ojas Nimase, Zhe Chen, Gengpei Qi, Yue Zhao, Xiyang Hu
GEO-Bench: Benchmarking Ranking Manipulation in Generative Engine Optimization
**arXiv preprint arXiv:2605.29107**

118. Shawn Li, Chenxiao Yu, Han Wang, Wei Yang, Ryan Rossi, Franck Dernoncourt, Xiyang Hu, Philip Yu, Chaowei Xiao, Huan Zhang, Yue Zhao
FORTIS: Benchmarking Over-Privilege in Agent Skills
**arXiv preprint arXiv:2605.09163**

117. Tiankai Yang, Jiate Li, Yi Nian, Shen Dong, Ruiyao Xu, Ryan Rossi, Kaize Ding, Yue Zhao
No Attacker Needed: Unintentional Cross-User Contamination in Shared-State LLM Agents
**arXiv preprint arXiv:2604.01350**

116. Haiyue Zhang, Yi Nian, Yue Zhao
Agent Audit: A Security Analysis System for LLM Agent Applications
**arXiv preprint arXiv:2603.22853**

115. Yi Nian, Haosen Cao, Shenzhe Zhu, Henry Peng Zou, Qingqing Luan, Yudi Zhang, Yue Zhao
When Only the Final Text Survives: Implicit Execution Tracing for Multi-Agent Auditing
**Under submission**
**arXiv preprint arXiv:2603.17445**

114. Aojie Yuan, Zhiyuan Su, Yue Zhao
AEGIS: No Tool Call Left Unchecked – A Pre-Execution Firewall and Audit Layer for AI Agents
**Under submission**
**arXiv preprint arXiv:2603.12621**

113. Rujie Ye, Jiayi Zhang, Zhuoxin Liu, Zihao Zhu, Siyuan Yang, Li Li, Tianfu Fu, Franck Dernoncourt, Yue Zhao, Jiacheng Zhu, Ryan Rossi, Wenhao Chai, Zhengzhong Tu
Agent Banana: High-Fidelity Image Editing with Agentic Thinking and Tooling
**Under submission**
**arXiv preprint arXiv:2602.09084**

112. Xiaolin Zhou, Zheng Luo, Yicheng Gao, Qixuan Chen, Xiyang Hu, Yue Zhao, Ruishan Liu
Fairness or Fluency? An Investigation into Language Bias of Pairwise LLM-as-a-Judge
**Under submission**
**arXiv preprint arXiv:2601.13649**

111. Kay Liu, Yuwei Han, Haoyan Xu, Henry Peng Zou, Yue Zhao, Philip S. Yu
TAGFN: A Text-Attributed Graph Dataset for Fake News Detection in the Age of LLMs
**Under submission**
**arXiv preprint arXiv:2511.21624**

110. Haoyan Xu, Ruizhi Qian, Zhengtao Yao, Ziyi Liu, Li Li, Yuqi Li, Yanshu Li, Wenqing Zheng, Daniele Rosa, Daniel Barcklow, Senthil Kumar, Jieyu Zhao, Yue Zhao
LLM-Powered Text-Attributed Graph Anomaly Detection via Retrieval-Augmented Reasoning
**Under submission**
**arXiv preprint arXiv:2511.17584**

109. Haoyan Xu, Ruizhi Qian, Jiate Li, Yushun Dong, Minghao Lin, Hanson Yan, Zhengtao Yao, Qinghua Liu, Junhao Dong, Ruopeng Huang, Yue Zhao, Mengyuan Li
A Systematic Study of Model Extraction Attacks on Graph Foundation Models
**Under submission**
**arXiv preprint arXiv:2511.11912**

108. Yuexing Hao, Yue Huang, Haoran Zhang, Chenyang Zhao, Zhenwen Liang, Paul Pu Liang, Yue Zhao, Lichao Sun, Saleh Kalantari, Xiangliang Zhang, Marzyeh Ghassemi
The Role of Computing Resources in Publishing Foundation Model Research
**Under submission**
**arXiv preprint arXiv:2510.13621**

107. Wang Wei, Tiankai Yang, Hongjie Chen, Yue Zhao, Franck Dernoncourt, Ryan A. Rossi, Hoda Eldardiry
Learning to Route LLMs from Bandit Feedback: One Policy, Many Trade-offs
**Under submission**
**arXiv preprint arXiv:2510.07429**

106. Langzhou He, Junyou Zhu, Fangxin Wang, Junhua Liu, Haoyan Xu, Yue Zhao, Philip S. Yu, Qitian Wu
Can Molecular Foundation Models Know What They Don't Know? A Simple Remedy with Preference Optimization
**Under submission**
**arXiv preprint arXiv:2509.25509**

105. Haoyan Xu, Zhengtao Yao, Xuzhi Zhang, Ziyi Wang, Langzhou He, Yushun Dong, Philip S. Yu, Mengyuan Li, Yue Zhao
GLIP-OOD: Zero-Shot Graph OOD Detection with Foundation Model
**Under submission**
**arXiv preprint arXiv:2504.21186**

104. Haoyan Xu, Zhengtao Yao, Ziyi Wang, Zhan Cheng, Xiyang Hu, Mengyuan Li, Yue Zhao
Graph Synthetic Out-of-Distribution Exposure with Large Language Models
**Under submission**
**arXiv preprint arXiv:2504.21198**

103. Kaixiang Zhao, Lincan Li, Kaize Ding, Neil Zhenqiang Gong, Yue Zhao, Yushun Dong
A Survey of Model Extraction Attacks and Defenses in Distributed Computing Environments
**Under submission**
**arXiv preprint arXiv:2502.16065**

102. Shixuan Li, Wei Yang, Peiyu Zhang, Xiongye Xiao, Defu Cao, Yuehan Qin, Xiaole Zhang, Yue Zhao, Paul Bogdan
ClimateLLM: Efficient Weather Forecasting via Frequency-Aware Large Language Models
**Under submission**
**arXiv preprint arXiv:2502.11059**

101. Lincan Li, Jiaqi Li, Catherine Chen, Fred Gui, other collaborators, Yue Zhao, Yushun Dong
Political-LLM: Large Language Models in Political Science
**Under submission**
**arXiv preprint arXiv:2412.06864**

100. Chenxiao Yu, Jinyi Ye, Yuangang Li, Zhaotian Weng, Zheng Li, Emilio Ferrara, Xiyang Hu, Yue Zhao
A Large-Scale Simulation on Large Language Models for Decision-Making in Political Science
**Under submission**
**arXiv preprint arXiv:2412.15291**

99. Junda Wu, Hanjia Lyu, Yu Xia, Zhehao Zhang, Joe Barrow, Ishita Kumar, Mehnoosh Mirtahebi, Hongjie Chen, Ryan A. Rossi, Franck Dernoncourt, Tong Yu, Ruiyi Zhang, Jiuxiang Gu, Nesreen K. Ahmed, Yu Wang, Xiang Chen, Hanieh Deilamsalehy, Namyong Park, Sungchul Kim, Huanrui Yang, Subrata Mitra, Zhengmian Hu, Nedim Lipka, Yue Zhao, Jiebo Luo, Julian McAuley
Personalized Multimodal Large Language Models: A Survey
**Under submission**
**arXiv preprint arXiv:2412.02142**

## Peer-reviewed Journal Papers

98. Yuehan Qin, Li Li, Defu Cao, Tiankai Yang, Jiate Li, Yue Zhao
M3OOD: Automatic Selection of Multimodal OOD Detectors
*IEEE Transactions on Neural Networks and Learning Systems (**TNNLS**)*, 2026

97. Jiechao Gao†, Yuangang Li†, Jie Wang, Michael Lepech, Yue Zhao, Brad Campbell
H-FedSN: Personalized Sparse Networks for Efficient and Accurate Hierarchical Federated Learning for IoT Applications
*npj Artificial Intelligence*, 2026.

96. Yuehan Qin, Shawn Li, Yi Nian, Xinyan Velocity Yu, Yue Zhao, Xuezhe Ma
Don't Let It Hallucinate: Premise Verification via Retrieval-Augmented Logical Reasoning
*Transactions on Machine Learning Research (TMLR)*, 2026

95. Haoyan Xu, Kay Liu, Zhengtao Yao, Philip S. Yu, Mengyuan Li, Kaize Ding, Yue Zhao
LEGO-Learn: Label-Efficient Graph Open-Set Learning
*Transactions on Machine Learning Research (**TMLR**)*, 2025

94. Hao Dong, Gaetan Frusque, Yue Zhao, Eleni Chatzi, Olga Fink
NNG-Mix: Improving Semi-supervised Anomaly Detection with Pseudo-anomaly Generation
*IEEE Transactions on Neural Networks and Learning Systems (**TNNLS**)*, 2024

93. Ling Yang†, Zhilong Zhang†, Yang Song, Shenda Hong, Runsheng Xu, Yue Zhao, Wentao Zhang, Bin Cui, Ming-Hsuan Yang
Diffusion Models: A Comprehensive Survey of Methods and Applications
*ACM Computing Surveys (**CSUR**)*, 2023

92. Yue Zhao†, Martin Q. Ma†, Xiaorong Zhang, Leman Akoglu
The Need for Unsupervised Outlier Model Selection: A Review and Evaluation of Internal Evaluation Strategies
*ACM SIGKDD Explorations Newsletter (**SIGKDD Explor.**)*, 2023

91. Kexin Huang†, Tianfan Fu†, Wenhao Gao†, Yue Zhao, Yusuf Roohani, Jure Leskovec, Connor W. Coley, Cao Xiao, Jimeng Sun, Marinka Zitnik
Artificial Intelligence Foundation for Therapeutic Science
*Nature Chemical Biology (**NCHEMB**)*, 2022

90. Zheng Li†, Yue Zhao†, Xiyang Hu, Nicola Botta, Cezar Ionescu, George H. Chen
ECOD: Unsupervised Outlier Detection Using Empirical Cumulative Distribution Functions
*IEEE Transactions on Knowledge and Data Engineering (**TKDE**)*, 2022.

89. Yue Zhao, Zain Nasrullah, Zheng Li
PyOD: A Python Toolbox for Scalable Outlier Detection
*Journal of Machine Learning Research* (**JMLR**), 2019.

## Conference & Workshop Papers

88. Zixiang Xu, Zirui Song, Chiyu Zhang, Xiuying Chen, Xi Liu, Xiyang Hu, Yue Zhao
JevOut: Natural Context Can Flip Decision Models
*NeurIPS 2026 Agenthon Workshop*, Poster.
arXiv:2609.30243.

87. Wei Yang, Shawn Li, Yuehan Qin, Yawei Wang, Mingxi Wang, Shixuan Li, Tiankai Yang, Jiate Li, Jesse Thomason, Xuezhe Ma, Yue Zhao
SynCo: Data Synthesis Co-Training for Self-Evolving LLMs via Multi-Agent Reinforcement Learning
*NeurIPS 2026 Workshop on Towards Test-Time Continual Learning Agents (TTCL)*, Poster.
arXiv:2610.11345.

86. Tiankai Yang, Yi Nian, Xinyuan Li, Ruiyao Xu, Henry Peng Zou, Kaize Ding, Xiyang Hu, Yan Liu, Yue Zhao
Cat-DPO: Category-Adaptive Safety Alignment
*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2026.

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83. Chenxiao Yu, Bowen Yi, Farzan Karimi-Malekabadi, Suhaib Abdurahman, Jinyi Ye, Shrikanth Narayanan, Yue Zhao, Morteza Dehghani
Tracing Moral Foundations in Large Language Models
*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2026.

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MaskForge: Structure-Aware Adaptive Attacks for Jailbreaking Diffusion Large Language Models
*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2026.

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*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2026.

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CHI-Bench: Can AI Agents Automate End-to-End, Long-Horizon, Policy-Rich Healthcare Workflows?
*Advances in Neural Information Processing Systems (**NeurIPS** Evaluations and Datasets Track)*, 2026.

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The Autonomy Tax: Defense Training Breaks LLM Agents
*Conference on Empirical Methods in Natural Language Processing (**EMNLP**)*, 2026.

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SpecAlign: Efficient Specification-Grounded Alignment of Large Language Models via Synthetic Data
*Conference on Empirical Methods in Natural Language Processing (**EMNLP**)*, 2026.

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76. Prince Zizhuang Wang, Aojie Yuan, Haiyue Zhang, Xiyang Hu, Yue Zhao, Shuli Jiang
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***Findings** of the Association for Computational Linguistics: **EMNLP***, 2026. Earlier versions presented at *SocialSim @ COLM 2025* (**Spotlight**) and *ResponsibleFM @ NeurIPS 2025*.

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***Findings** of the Association for Computational Linguistics: **EMNLP***, 2026.

73. Aojie Yuan, Yi Nian, Haiyue Zhang, Zijian Su, Yue Zhao
SEVA: Self-Evolving Verification Agent with Process Reward for Fact Attribution
***ICML** 2026 **Workshops** (**AI4GOOD**; **FAGEN**).*

72. Wang Wei, Hongzheng Yang, Tiankai Yang, Samyadeep Basu, Hongjie Chen, Yue Zhao, Franck Dernoncourt, Ryan A. Rossi, Hoda Eldardiry
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*Conference on Language Modeling (**COLM**)*, 2026.

71. Yiming Tang, Yi Fan, Chenxiao Yu, Tiankai Yang, Yue Zhao, Xiyang Hu
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***ICML Workshop** on Trustworthy AI for Good (**AI4GOOD**)*, 2026.

70. Han Bao, Yue Huang, Yanbo Wang, Jiayi Ye, Xiangqi Wang, Xiuying Chen, Yue Zhao, Tianyi Zhou, Mohamed Elhoseiny, Xiangliang Zhang
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*ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD** Datasets and Benchmarks Track)*, 2026.

69. Zhisheng Qi, Utkarsh Sahu, Li Ma, Haoyu Han, Ryan Rossi, Franck Dernoncourt, Mahantesh Halappanavar, Nesreen Ahmed, Yushun Dong, Yue Zhao, Yu Zhang, Yu Wang
Benchmarking Knowledge-Extraction Attack and Defense on Retrieval-Augmented Generation
*ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD** Datasets and Benchmarks Track)*, **Oral**, 2026.

68. Jiate Li, Defu Cao, Li Li, Wei Yang, Yuehan Qin, Chenxiao Yu, Tiannuo Yang, Ryan A. Rossi, Yan Liu, Xiyang Hu, Yue Zhao
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67. Shawn Li, Chenxiao Yu, Zhiyu Ni, Hao Li, Charith Peris, Chaowei Xiao, Yue Zhao
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*Annual Meeting of the Association for Computational Linguistics (**ACL**)*, 2026.

66. Ruiyao Xu, Mihir Parmar, Tiankai Yang, Zhengyu Hu, Yue Zhao, Kaize Ding
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65. Jinbo Liu, Defu Cao, Yifei Wei, Tianyao Su, Yuan Liang, Yushun Dong, Yan Liu, Yue Zhao, Xiyang Hu
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61. Weidi Luo, Tianyu Lu, Qiming Zhang, Xiaogeng Liu, Bin Hu, Yue Zhao, Jieyu Zhao, Song Gao, Patrick McDaniel, Zhen Xiang, Chaowei Xiao
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*International Conference on Learning Representations (**ICLR**)*, 2026

60. Yue Huang, Chujie Gao, Siyuan Wu, Haoran Wang, Xiangqi Wang, Yujun Zhou, Yanbo Wang, Jiayi Ye, Jiawen Shi, Qihui Zhang, Yuan Li, Han Bao, Zhaoyi Liu, Tianrui Guan, Dongping Chen, Ruoxi Chen, other authors, Yue Zhao, other authors, Xiangliang Zhang
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*The Web Conference (**WWW**)*, 2026

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*Conference of the European Chapter of the Association for Computational Linguistics (**EACL**)*, 2026

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*Annual Meeting of the Association for Computational Linguistics (**ACL**)*, 2025.

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*ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD** Lecture-Style Tutorial Track)*, 2025.

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*Conference on Computer Vision and Pattern Recognition (**CVPR**)*, 🎗 **Highlight**, 2025

38. Hanhui Wang, Yihua Zhang, Ruizheng Bai, Yue Zhao, Sijia Liu, Zhengzhong Tu
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*Conference on Computer Vision and Pattern Recognition (**CVPR**)*, 2025

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*Annual Conference of the North American Chapter of the Association for Computational Linguistics (**NAACL** Demo Track)*, 2025.

36. Yuehan Qin†, Yichi Zhang†, Yi Nian†, Xueying Ding, Yue Zhao
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*International Conference on Learning Representations (**ICLR**)*, 2025
*ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (***KDD Workshop** on Resource-Efficient Learning for Knowledge Discovery*), 🎗 Best Paper Award, 2024.

35. Sihan Chen, Zhuangzhuang Qian, Wingchun Siu, Xingcan Hu, Jiaqi Li, Shawn Li, Yuehan Qin, Tiankai Yang, Zhuo Xiao, Wanghao Ye, Yichi Zhang, Yushun Dong, Yue Zhao
PyOD 2: A Python Library for Outlier Detection with LLM-powered Model Selection
*International World Wide Web Conference (**The Web Conference** Demo Track)*, 2025

34. Sizhe Liu, Yizhou Lu, Siyu Chen, Xiyang Hu, Jieyu Zhao, Tianfan Fu, Yue Zhao
DrugAgent: Automating AI-aided Drug Discovery Programming through LLM Multi-Agent Collaboration
***AAAI Workshop** on Foundation Models for Biological Discoveries (**FMs4Bio**)*, 2025.

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*Advances in Neural Information Processing Systems (**NeurIPS**)*, 🎗 **Spotlight**, 2024

32. Xueying Ding, Yue Zhao, Leman Akoglu
Fast Unsupervised Deep Outlier Model Selection with Hypernetworks
*ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2024*

31. Lichao Sun, Yue Huang, Haoran Wang, Siyuan Wu, Qihui Zhang, Chujie Gao, Yixin Huang, Wenhan Lyu, Yixuan Zhang, Xiner Li, Zhengliang Liu, Yixin Liu, Yijue Wang, Zhikun Zhang, 50+ collaborative authors, Yue Zhao
TrustLLM: Trustworthiness in Large Language Models
*International Conference on Machine Learning (**ICML**)*, 2024

30. Yue Zhao, Leman Akoglu
HPOD: Hyperparameter Optimization for Unsupervised Outlier Detection
*International Conference on Automated Machine Learning (**AutoML**)*, 2024

29. Yue Zhao
Towards Reproducible, Automated, and Scalable Anomaly Detection
*AAAI Conference on Artificial Intelligence (**AAAI**), New Faculty Highlights*, 2024

28. Minqi Jiang[†], Chaochuan Hou[†], Ao Zheng[†], Songqiao Han, Hailiang Huang, Qingsong Wen, Xiyang Hu, Yue Zhao
ADGym: Design Choices for Deep Anomaly Detection.
*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2023

27. Jaemin Yoo, Yue Zhao, Lingxiao Zhao, Leman Akoglu
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*European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (**ECML/PKDD**)*, 2023

26. Peng Xu, Lin Zhang, Xuanzhou Liu, Jiaqi Sun, Yue Zhao, Haiqin Yang, Bei Yu
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*International Conference on Machine Learning (**ICML**)*, 2023

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ADMoE: Anomaly Detection with Mixture-of-Experts from Noisy Labels
*Thirty-Seventh AAAI Conference on Artificial Intelligence (**AAAI**)*, 2023

24. Yue Zhao, George H. Chen, Zhihao Jia
TOD: GPU-accelerated Outlier Detection via Tensor Operations
*International Conference on Very Large Data Bases (**VLDB**)*, 2023

23. Songqiao Han[†], Xiyang Hu[†], Hailiang Huang[†], Minqi Jiang[†], Yue Zhao[†]
ADBench: Anomaly Detection Benchmark
*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2022
([†]Equal contribution)

22. Yue Zhao[†], Kay Liu[†], Yingtong Dou[†], et al.
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*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2022.

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Toward Unsupervised Outlier Model Selection
*IEEE International Conference on Data Mining (**ICDM**)*, 2022.

20. Zhiming Xu, Xiao Huang, Yue Zhao, Yushun Dong, Jundong Li
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*Pacific-Asia Conference on Knowledge Discovery and Data Mining (**PAKDD**)*, 2022.

19. Yue Zhao, Ryan A. Rossi, Leman Akoglu
Automatic Unsupervised Outlier Model Selection
*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2021.

18. Kwei-Herng Lai, Daochen Zha, Junjie Xu, Yue Zhao, Guanchu Wang, Xia Hu
Revisiting Time Series Outlier Detection: Definitions and Benchmarks
*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2021

17. Kexin Huang[†], Tianfan Fu[†], Wenhao Gao[†], Yue Zhao, Yusuf Roohani, Jure Leskovec, Connor W. Coley, Cao Xiao, Jimeng Sun, Marinka Zitnik
Therapeutics Data Commons: Machine Learning Datasets and Tasks for Drug Discovery and Development
*Advances in Neural Information Processing Systems (**NeurIPS**)*, 2021

16. Yue Zhao[†], Xiyang Hu[†], Cheng Cheng, Cong Wang, Changlin Wan, Wen Wang, Jianing Yang, Haoping Bai, Zheng Li, Cao Xiao, Yunlong Wang, Zhi Qiao, Jimeng Sun, Leman Akoglu
SUOD: Accelerating Large-scale Unsupervised Heterogeneous Outlier Detection
*Conference on Machine and Learning Systems (**MLSys**)*, 2021.

15. Kwei-Herng Lai†, Daochen Zha†, Guanchu Wang, Junjie Xu, Yue Zhao, Devesh Kumar, Yile Chen, Purav Zumkhawaka, Minyang Wan, Diego Martinez, Xia Hu
TODS: An Automated Time Series Outlier Detection System (Demo paper)
*Thirty-Fifth AAAI Conference on Artificial Intelligence (**AAAI**)*, 2021.

14. Meng-Chieh Lee, Yue Zhao, Aluna Wang, Pierre Jinghong Liang, Leman Akoglu, Vincent S. Tseng, Christos Faloutsos
AutoAudit: Mining Accounting and Time-Evolving Graphs
*IEEE International Conference on Big Data (**Big Data**)*, 2020

13. Changlin Wan, Dongya Jia, Yue Zhao, Wennan Chang, Sha Cao, Xiao Wang, and Chi Zhang
A Data Denoising Approach to Optimize Functional Clustering of Single Cell RNA-sequencing Data
*IEEE International Conference on Bioinformatics and Biomedicine (**BIBM**)*, 2020

12. Yue Zhao, Xueying Ding, Jianing Yang, Haoping Bai.
SUOD: Toward Scalable Unsupervised Outlier Detection
***Workshops** at the Thirty-Fourth **AAAI** Conference on Artificial Intelligence*, 2020.
**Extended version published in *MLSys 2021*.**

11. Zheng Li, Yue Zhao, Nicola Botta, Cezar Ionescu, Xiyang Hu
COPOD: Copula-Based Outlier Detection
*IEEE International Conference on Data Mining (**ICDM**)*, 2020.

10. Zheng Li, Yue Zhao, Jialin Fu
SYNC: A Copula based Framework for Generating Synthetic Data from Aggregated Sources
*IEEE International Conference on Data Mining Workshops (**ICDMW**)*, 2020.

9. Yiqun Mei, Yue Zhao, Wei Liang
DSR: An Accurate Single Image Super Resolution Approach for Various Degradations
*IEEE International Conference on Multimedia and Expo (**ICME**)*, 2020, London, UK.

8. Yue Zhao, Xuejian Wang†, Cheng Cheng†, Xueying Ding†
Combining Machine Learning Models and Scores using combo Library (Demo paper)
*Thirty-Fourth AAAI Conference on Artificial Intelligence (**AAAI**)*, 2020.

7. Colin Wan, Zheng Li, Alicia Guo, Yue Zhao
SynC: A Unified Framework for Generating Synthetic Population with Gaussian Copula
***Workshops** at the Thirty-Fourth **AAAI** Conference on Artificial Intelligence*, 2020.
**Extended version published in *ICDMW 2020*.**

6. Zain Nasrullah, Yue Zhao
Music Artist Classification with Convolutional Recurrent Neural Networks
*IEEE International Joint Conference on Neural Networks* (***IJCNN***), 2019, Hungary.

5. Yue Zhao, Zain Nasrullah, Maciej K. Hryniewicki, Zheng Li
LSCP: Locally Selective Combination in Parallel Outlier Ensembles
*SIAM International Conference on Data Mining* (***SDM***), 2019, Calgary, Canada.

4. Yue Zhao, Maciej K. Hryniewicki
DCSO: Dynamic Combination of Detector Scores for Outlier Ensembles
*ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (***KDD Workshop** on Outlier Detection De-constructed*), 2018, London, UK.
**Extended version published in *SDM 2019*, renamed to LSCP.**

3. Yue Zhao, Maciej K. Hryniewicki
XGBOD: Improving Supervised Outlier Detection with Unsupervised Representation Learning
*IEEE International Joint Conference on Neural Networks* (***IJCNN***), 2018, Rio, Brazil.

2. Yue Zhao, Maciej K. Hryniewicki, Francesca Cheng, Boyang Fu, Xiaoyu Zhu
Employee Turnover Prediction with Machine Learning: A Reliable Approach
*Intelligent System Conference* (***Intellisys***), 2018, London, UK.

1. Yue Zhao†, Zhongtian Qiu†, Yiqing Yang†, Weiwei Li†, Mingming Fan
An Empirical Study of Touch-based Authentication Methods on Smartwatches
*ACM International Symposium on Wearable Computers* (***ISWC***), 2017, Maui, USA.

COMMENTARY

Angela Huyue Zhang and Yue Zhao. Independent AI Auditing Now. *Project Syndicate*, October 15, 2026.

## Internship Experience

**NortonLifeLock Research Group**
Machine Learning Research Intern — 2022

**Microsoft Research**
Machine Learning Research Intern — 2022

**Stanford University, Computer Science Department**
Visiting Student Researcher (Prof. Jure Leskovec) — 2021

**IQVIA, Analytics Center of Excellence**
Machine Learning Research Intern — 2020

**Siemens PLM Software USA**
Software Engineer (Intern & Contract) — Mar. 2012 - Dec. 2014

## Teaching Experience

***University of Southern California*** — Los Angeles, CA

**Instructor** — Spring 2027 (scheduled)
*CSCI 699 Auditing AI Systems: Foundation Models and Agents*

**Instructor** — Fall 2026 (scheduled)
*CSCI 566 Deep Learning and Its Applications*

**Instructor** — Spring 2026
*CSCI 699 Adversarial and Trustworthy Foundation Models*

**Instructor** — Spring 2025
*CSCI 566 Deep Learning and Its Applications*

**Instructor** — Spring 2024
*CSCI 566 Deep Learning and Its Applications*

***Carnegie Mellon University*** — Pittsburgh, PA

**Teaching Assistant** — Fall 2022
*Managing Digital Business* (Prof. David Riel)

**Teaching Assistant & co-Instructor** (lectures on AutoML and MLSys) — Spring 2022 – Fall 2020
*Intro to Artificial Intelligence* (Prof. David Steier)

**Teaching Assistant** — Spring 2022
*Digital Transformation* (Prof. David Riel)

**Teaching Assistant** (helping on course topics) — Fall 2021
*Statistics for IT Managers* (Prof. Daniel Nagin)

***University of Toronto*** — Toronto, ON

**Teaching Assistant & Lab Session Instructor** — Fall 2015
*Embedded Systems* (Prof. Philip Anderson)

***University of Cincinnati*** — Cincinnati, OH

**Teaching Assistant & Lab Session Instructor** — Fall 2014
*Intro to Programming* (Prof. George Purdy)

## Ph.D. Students

**Current Students**

- Yuehan Qin (USC, CS Ph.D. Candidate, 2024 Fall-)
- Tiankai Yang (USC, CS Ph.D., 2024 Fall-)
- Shawn Li (USC, CS Ph.D. Candidate, 2024 Fall-), Capital One Fellowship, Amazon ML Fellowship
- Jiate Li (USC, CS Ph.D., 2025 Fall-)
- Wei Yang (USC, CS Ph.D. Candidate, 2026 Fall-), co-advised by Jesse Thomason and Xuezhe Ma
- Yi Nian (USC, CS Ph.D., incoming)
- Chenxiao Yu (USC, CS Ph.D., 2026 Fall-)
- Zixiang Xu (USC, CS Ph.D., 2026 Fall-)

### Alumni

- Haoyan Xu (USC, ECE Ph.D., advised 2024 Spring–2026 Oct), co-advised by Mengyuan Li, Capital One Fellowship; passed Ph.D. defense Oct. 2026; destination: Microsoft

## Student Committee

| Term | Student | Committee |
|---|---|---|
| Fall 2026 | Chen Chu (USC, CS Ph.D.) | Qualification |
| | Yuqing Yang (USC, CS Ph.D.) | Qualification |
| | Zeyu Shangguan (USC, CS Ph.D.) | Qualification |
| | Chufan Shi (USC, CS Ph.D.) | Qualification |
| | Linxin Song (USC, CS Ph.D.) | Qualification |
| | Raja Kumar (USC, CS Ph.D.) | Qualification |
| | Ziyi Liu (USC, CS Ph.D.) | Thesis |
| Summer 2026 | Defu Cao (USC, CS Ph.D.) | Defense |
| Spring 2026 | Longchao Da (ASU, CS Ph.D.) | Defense |
| | Yahan Li (USC, CS Ph.D.) | Qualification |
| | Jiageng Mao (USC, CS Ph.D.) | Thesis |
| | Xinyue Cui (USC, CS Ph.D.) | Qualification |
| | Jiawei Yang (USC, CS Ph.D.) | Qualification + Thesis |
| | Elnaz Rahmati (USC, CS Ph.D.) | Qualification |
| Fall 2025 | Hao Dong (ETH Zurich, CS Ph.D.) | Defense |
| | Haonan Wang (USC, ECE Ph.D.) | Defense |
| Spring 2025 | Mengxi Wu (USC, CS Ph.D.) | Qualification |
| | Xingrui Wang (USC, ME Ph.D.) | Defense |
| Fall 2024 | Alex Bisberg (USC, CS Ph.D.) | Thesis |
| Summer 2024 | Gengyu Rao (USC, CS Ph.D.) | Defense |
| | Mehrdad Kiamari (USC, ECE Ph.D.) | Defense |
| Spring 2024 | Maria Despoina Siampou (USC, CS Ph.D.) | Qualification |
| | Yuan Meng (USC, ECE Ph.D.) | Defense |
| | Hassan Hamad (USC, ECE Ph.D.) | Qualification |
| | Yizhou Zhang (USC, CS Ph.D.) | Thesis |
| | Haoming Li (USC, CS Ph.D.) | Qualification |
| | Arash Hajisafi (USC, CS Ph.D.) | Qualification |
| Fall 2023 | Yi Chien Lin (USC, ECE Ph.D.) | Qualification |
| | Yuke Zhang (USC, ECE Ph.D.) | Qualification |

## Services

### Institutional Leadership

- Associate Co-Director, USC Institute on Ethics and Trust in Computing (IETC), 2026–2027

### Community Outreach

- Founder & Maintainer, CSPhD.org: a non-profit community for CS/AI/EE/Stats PhDs (since 2022, several thousand members, 500+ PhD/RA/postdoc openings), supporting students from before enrollment to the faculty job market.

### Conference/Workshop Organizing Roles

- Co-organizer, Geometric and Spatial Reasoning in Vision-Language Models (GSR-VLM) Workshop @ AAAI 2027
- Doctoral Consortium Co-Chair, AAAI 2027
- Future Faculty Highlights Co-Chair, IEEE BigData 2026
- Publicity Co-Chair, WSDM 2027
- Workflow Co-Chair, KDD 2023

- Co-organizer, SURGeLLM: Structured Understanding, Retrieval, and Generation in LLMs era Workshop @ ACL 2026
- Co-organizer, AgentSkills '26: The First Workshop on Agent Skills @ CAIS Conference 2026
- Co-organizer, AI for Financial Fraud Detection & Prevention Workshop @ 6th ACM International Conference on AI in Finance

**Funding Proposal Review Roles**

- National Science Foundation (NSF)
- Dutch Research Council (NWO)

**Journal Editorial Roles**

- Action Editor, Transactions on Machine Learning Research (TMLR), 2026–present
- Associate Editor, ACM Transactions on AI for Science (TAIS), 2025–present
- Associate Editor, IEEE Transactions on Neural Networks and Learning Systems (TNNLS), 2024–present
- Action Editor, Journal of Data-centric Machine Learning Research (DMLR), 2024–present

**Conference/Workshop AC and Reviewer Roles**

- ICLR 2025 (AC), ICLR 2026 (AC)
- AAAI 2021, 2022, 2023, 2025 (Senior PC), 2026 (Senior PC)
- ICML 2024, 2025 (AC), 2026 (AC)
- NeurIPS 2021, 2022, 2023, 2025 (AC), 2026 (AC)
- AISTATS 2024, 2025 (AC)
- MLSys 2024, 2026
- KDD 2020, 2021, 2022, 2023, 2026
- ACL Rolling Review (ARR) 2026
- IJCAI 2022, 2023
- AAAI Demonstrations 2021, 2022
- MICCAI 2020, 2021, 2022
- ICDM 2020
- KDD Workshop on Outlier Detection and Description (ODD), 2021
- KDD Workshop on Anomaly and Novelty Detection (ANDEA), 2021, 2022
- IJCAI Workshop on Artificial Intelligence for Anomalies and Novelties (AI4AN), 2020, 2021
- INFORMS Workshop on Data Science 2021

**Journal Review Roles**

- Journal of Machine Learning Research (JMLR)
- PNAS Nexus
- Machine Learning
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- IEEE Internet of Things Journal (IoT-J)
- IEEE Intelligent Systems
- IEEE Journal on Selected Areas in Communications (J-SAC)
- Data Mining and Knowledge Discovery (DMAI)
- ACM Transactions on Management Information Systems (TMIS)
- Knowledge and Information Systems (KAIS)
- INFORMS Journal on Computing (IJOC)
- Big Data
- Artificial Intelligence Review (AIRE)

- Neurocomputing
- IEEE Transactions on Systems, Man, and Cybernetics: Systems
- IEEE/ACM Transactions on Computational Biology and Bioinformatics (TCBB)
- IEEE Network Magazine
- IEEE Computational Intelligence Magazine (CIM)
- BioData Mining
- European Journal of Management and Business Economics (EJMBE)
- The Journal of Open Source Software (JOSS)

## Talks and Lectures

| | | |
|---|---|---|
| ASTAD @ AAAI 2027 | **Invited speaker (upcoming)**. 4th Workshop on Automated Spatial and Temporal Anomaly Detection | Feb. 2027 |
| Trust4RSI-Agent @ AAAI 2027 | **Invited speaker (upcoming)**. Workshop on Trustworthy Evolution of AI Agents | Feb. 2027 |
| USC IETC Reading Group | *Safe Models, Unsafe Agents: From Defense Training to Agent Auditing* | Sep. 2026 |
| Arizona State University | **Guest Lecture** *My Unexpected Programming Journey* | Sep. 2026 |
| NVIDIA Agent & Security | *Safe Models, Broken Agents: How Deployed LLM Agents Break Without an Attacker* | May. 2026 |
| Amazon AI Security | *Safe Models, Broken Agents: How Deployed LLM Agents Break Without an Attacker* | May. 2026 |
| USC symposium on Frontiers of ML/AI | *Towards Robust AI: Advances in Outlier and OOD Detection* | Mar. 2025 |
| NUS Tea Talk | *Towards Robust AI: Advances in Outlier and OOD Detection* | Jan. 2025 |
| SFU@NeurIPS'24 | *Towards Robust AI: Advances in Outlier and OOD Detection* | Dec. 2024 |
| KAIST | *Unsupervised Model Selection: Automation with Meta-learning and LLMs* | Nov. 2024 |
| Kennesaw State University | *Unsupervised Model Selection: Automation with Meta-learning and LLMs* | Oct. 2024 |
| LinkedIn Anti-Abuse AI | *Outlier Detection: Automation, Systems, and GenAI* | Aug. 2024 |
| Amazon Security AI | *Outlier Detection: Automation, Systems, and GenAI* | Aug. 2024 |
| New York University | *Outlier Detection: Automation, Systems, and GenAI* | Aug. 2024 |
| University of Washington | *Outlier Detection: Automation, Systems, and GenAI* | Jun. 2024 |
| Microsoft | *Outlier Detection: Automation, Systems, and GenAI* | Jun. 2024 |
| USC Retreat on AI and Engineering Safety | *Safety Measures for LLMs* | Apr. 2024 |
| Visa Research | *Towards Reproducible, Automated, and Scalable AD* | Apr. 2024 |
| USC Symposium on Frontiers of Generative AI | *Generative AI for Anomaly Detection* | Mar. 2024 |
| AAAI New Faculty Highlights (invited) | *Towards Reproducible, Automated, and Scalable AD* | Feb. 2024 |
| U of Nevada, Las Vegas | *Automated and Scalable ML Algorithms and Systems* | Oct. 2023 |
| Samsung Seminar | *Automated and Scalable Anomaly Detection Systems* | Aug. 2023 |
| KDD SoCal Day | *Enable Applications by ML with Noisy Inputs* | Aug. 2023 |
| CMU Catalyst | *How (Not) to Fail Your Academic Job Search* | May. 2023 |
| KAUST | *Automated and Scalable ML Algorithms and Systems* | Apr. 2023 |
| Emory University | *Automated and Scalable ML Algorithms and Systems* | Apr. 2023 |
| USC | *Automated and Scalable ML Algorithms and Systems* | Mar. 2023 |
| UC Davis | *Automated and Scalable ML Algorithms and Systems* | Mar. 2023 |
| Stony Brook University | *Automated and Scalable ML Algorithms and Systems* | Feb. 2023 |
| University of Chicago | *Automated and Scalable ML Algorithms and Systems* | Feb. 2023 |
| UC Merced | *Automated and Scalable ML Algorithms and Systems* | Feb. 2023 |
| CMU PDL Meeting | *Automated and Scalable ML Algorithms and Systems* | Jan. 2023 |
| CMU Data Science Seminar | **Guest Lecture** *Automated Anomaly Detection* | Nov. 2022 |
| LoG Seminar | *Large-scale Graph Anomaly Detection* | Oct. 2022 |
| Intuit | *Anomaly Detection for Financial Risk Modeling* | Aug. 2022 |
| Rice University | *Large-scale Anomaly Detection with Automation* | Sep. 2022 |
| Microsoft Research | *Weakly-supervised Anomaly Detection* | Sep. 2022 |
| Wells Fargo | *Anomaly Detection for Financial Risk Modeling* | Aug. 2022 |
| Columbia University | **Guest Lecture** *Anomaly Detection* | Jul. 2022 |
| Morgan Stanley | *Automated Outlier Detection* | Jun. 2022 |
| Microsoft Research | *Automated Outlier Detection* | Jun. 2022 |
| Morgan Stanley | *Large-scale Anomaly Detection Systems* | Mar. 2022 |
| Rutgers Business School | *Outlier Model Selection* | Mar. 2022 |
| Tesla | *Large-scale Anomaly Detection Systems* | Feb. 2022 |
| Catalyst, CMU | *Systems for Data Mining Algorithms* | Dec. 2021 |
| E&Y Canada | *ML applications in Data Analytics* | Oct. 2021 |
| University of Nottingham | *General Machine Learning Applications* | Jan. 2021 |